

OS Unit 5 — File Systems & I/O Complete Notes

Operating Systems (OS) · Unit 5 · File Systems & I/O · 2025

Overview

These notes cover **File Systems & I/O**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

1. File Concepts

Files have attributes, operations and access methods such as sequential and direct access.

Why it matters: File Concepts appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of file concepts in one line.
- A small example showing file concepts in practice.
- Common mistakes students make and how to avoid them.

2. Directory Structures

Single-level, two-level, tree and acyclic-graph directories organise files.

Why it matters: Directory Structures appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of directory structures in one line.
- A small example showing directory structures in practice.
- Common mistakes students make and how to avoid them.

3. Allocation Methods

Contiguous, linked and indexed allocation trade fragmentation against access speed.

Why it matters: Allocation Methods appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of allocation methods in one line.
- A small example showing allocation methods in practice.
- Common mistakes students make and how to avoid them.

4. Free Space Management

Bit vectors, linked lists and grouping track free disk blocks.

Why it matters: Free Space Management appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of free space management in one line.
- A small example showing free space management in practice.
- Common mistakes students make and how to avoid them.

5. Disk Scheduling

FCFS, SSTF, SCAN, C-SCAN and LOOK order disk requests to reduce seek time.

Why it matters: Disk Scheduling appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of disk scheduling in one line.
- A small example showing disk scheduling in practice.
- Common mistakes students make and how to avoid them.

6. RAID

RAID levels combine disks for performance and redundancy using striping, mirroring and parity.

Why it matters: RAID appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of raid in one line.
- A small example showing raid in practice.
- Common mistakes students make and how to avoid them.

Quick revision

[File Concepts](#) · [Directory Structures](#) · [Allocation Methods](#) · [Free Space Management](#) · [Disk Scheduling](#) · [RAID](#)